

HERALD COLLEGE KATHMANDU

SOFTWARE REQUIREMENTS
DOCUMENT

INTERN PERFORMANCE & ATTENDANCE TRACKING SYSTEM

Document Type	Software Requirements Document (SRD)
Project Name	InternTrack — Herald College
Prepared For	Herald College Kathmandu
Version	v1.0 — Initial Draft
Date	June 2026

1. Introduction

1.1 Purpose

This document defines the complete software requirements for InternTrack — a web-based intern performance and attendance tracking system for Herald College Kathmandu. It is intended for use by the development team, project manager, and college administration.

1.2 Project Background

Herald College currently manages 20–30 interns using an Excel-based attendance and reporting system. Interns perform the following core activities:

- Attending college on a scheduled shift time
- Making outbound calls to prospective students (call center activity)
- Conducting college tours for visiting students
- Working Saturday shifts (which generate Compensation Leave entitlements)

The current Excel system creates challenges in real-time visibility, accountability, performance tracking, and management oversight — especially when the manager is occupied in counselling sessions. InternTrack will solve these challenges through a structured web platform connected to the existing Excel data.

1.3 Scope

InternTrack will cover:

- Web-based daily attendance marking with status codes
- Real-time call and college tour performance logging
- Compensation Leave tracking (auto-calculated from Saturday attendance)
- Leave application and approval workflow
- Manager dashboard for oversight without disrupting counselling work
- Two-way sync with the existing Excel tracker
- Automated alerts and performance flags

1.4 Definitions

Term	Definition
P	Present — intern attended on time
A	Absent — did not attend without prior notice
L	Late — arrived after scheduled shift time
HD	Half Day — attended for only half of the shift

AL	Approved Leave — leave pre-approved by manager
UL	Unapproved Leave — absent without prior approval
CL	Compensation Leave — earned by working Saturday
ND	No Duty — intern had no Saturday roster that week; excluded from all calculations
College Tour	A guided visit conducted by an intern for prospective students visiting campus
Call Center Activity	Outbound calls made to prospective students to invite or follow up
KPI	Key Performance Indicator — measurable target set per intern

2. User Roles

The system will support three user roles, each with distinct permissions and views.

Role	Who	Key Permissions
Admin / Manager	Head of admissions / counselling manager	Full access — view all interns, approve leaves, set KPIs, export reports, override any entry
Supervisor	Senior intern or team lead	Mark attendance for team, log daily work output, view team dashboard, flag issues
Intern	Individual call center / tour intern	View own attendance, log own calls/tours, apply for leave, view own performance

3. Functional Requirements

3.1 Authentication & Onboarding

3.1.1 Login System

- Email and password login for all users
- Role-based access control — intern sees only own data; supervisor sees team; admin sees all
- Password reset via email OTP
- Session timeout after 30 minutes of inactivity

3.1.2 Intern Profile Management

- Admin can add, edit, deactivate intern profiles
- Profile fields: Full name, phone, email, joining date, shift schedule, role, assigned supervisor
- Saturday duty roster — admin marks which Saturdays each intern is rostered (determines ND vs P/A)

3.2 Attendance Tracking

3.2.1 Daily Attendance Grid

- Monthly calendar view — one row per intern, one column per date
- Each cell shows the day's status code: P, A, L, HD, AL, UL, CL, ND
- Supervisor or admin marks attendance — dropdown with all 8 options per cell
- Interns can also self-check-in via a Check-In button on their dashboard within the allowed shift window
- Late check-in auto-marks status as L if intern checks in after shift start time
- Colour coding matches Excel: Green=P, Red=A/UL, Yellow=L/HD/AL, Blue=CL, Grey=ND

3.2.2 Shift Time Tracking

- Each intern has an assigned shift start and end time stored in their profile
- System timestamps check-in and check-out
- Late arrival auto-calculated as difference between scheduled start and actual check-in
- If intern checks in more than 15 minutes late → status auto-suggests L (supervisor can override)

3.2.3 Saturday & Compensation Leave Logic

- Each Saturday, the system checks each intern's roster:
 - If rostered AND marks P → CL earned (valid Sun–Fri of the following week)
 - If rostered AND marks A/L → absence recorded, no CL earned
 - If NOT rostered → ND auto-applied, excluded from all performance and attendance calculations
- Comp Leave Tracker auto-populates when Saturday P is marked
- When intern marks CL on a following weekday → CL Used is auto-recorded

- System warns if intern tries to mark CL but has no available CL balance
- CL balance shown on intern dashboard at all times

3.3 Performance Tracking

3.3.1 Call Center Activity Log

- Intern logs calls made each day — either manually via a form or imported from Excel
- Fields per entry: Date, Calls Made, Calls Connected, Visits Booked, Registrations Confirmed
- Daily target visible on screen — auto-highlights if target not met (amber) or exceeded (green)
- Weekly and monthly call totals auto-calculated per intern
- Supervisor can add notes per intern per day

3.3.2 College Tour Log

- Intern logs each college tour conducted
- Fields: Date, Student Name (visitor), Number of Visitors, Outcome (Interested / Not Interested / Enrolled), Notes
- Monthly tour count per intern visible in summary
- Manager can see which interns are conducting tours and outcomes

3.3.3 KPI & Target Management

- Admin sets monthly KPI targets per intern or per role group:
 - Minimum calls per day
 - Minimum visits booked per week
 - Minimum college tours per month
 - Minimum attendance rate (%)
- System auto-flags interns below target with colour-coded status on dashboard

3.4 Leave Management

3.4.1 Leave Application

- Intern submits leave request from their dashboard: date, leave type (AL/HD/CL), reason
- System checks CL balance before allowing CL application
- Supervisor receives notification and can approve or reject
- Admin can override any leave decision
- Approved leave auto-updates the attendance grid

3.4.2 Leave Pattern Detection

- System detects and flags:
 - Same weekday repeated leaves (e.g., always absent Monday)
 - 3 or more leaves in a single month

- Unapproved leave more than once in a month
- CL used exceeding CL earned
- Flagged patterns shown on manager dashboard with intern name highlighted

3.5 Manager Dashboard

The manager dashboard is designed for use during brief gaps between counselling sessions — everything important is visible at a glance without clicking through multiple screens.

3.5.1 Daily Overview Panel

- Total interns present / absent / late today — updated in real time
- Who has not yet checked in (past shift start time)
- Calls made today across all interns (running total)
- College tours conducted today
- Any pending leave requests requiring approval

3.5.2 Performance Summary Panel

- Monthly leaderboard — interns ranked by calls made + tours conducted
- Red / Yellow / Green flag per intern based on KPI achievement
- CL balance overview — who has unused or overspent CL
- Attendance rate per intern for the current month

3.5.3 Alerts & Notifications

- Push notification (browser + email) for:
 - Intern checks in late
 - Intern marks absent without prior leave request
 - Leave pattern flag triggered
 - KPI below target for 3 consecutive days
 - Leave request pending for more than 24 hours

4. Excel Integration

4.1 Overview

The existing Excel tracker (Intern_Tracker_Final.xlsx) contains the attendance grid, comp leave tracker, and monthly summary for all 30 interns. The website must stay in sync with this file so that the manager can use either the website or the Excel sheet without data conflicts.

4.2 Sync Approach

Direction	Trigger	What Syncs
Excel → Website	Manual upload or scheduled (daily)	Attendance codes, call counts, leave records pulled from Excel into the web database
Website → Excel	On-demand export button	Full monthly attendance grid, summary, and comp leave tracker exported as formatted .xlsx matching the existing template
Real-time (future)	Auto-sync every 15 min	Two-way live sync using Google Sheets as the shared layer (Phase 2 option)

4.3 Excel Upload Specification

- Admin uploads .xlsx file via the website settings panel
- System reads sheets: "Monthly Attendance", "Comp Leave Tracker", "Daily Work Log"
- System maps intern names to existing website profiles by Intern ID (INT-01, INT-02...)
- Conflicts flagged for review — system shows row-by-row diff before confirming import
- Invalid codes or unknown intern IDs are listed in an error report before import proceeds

4.4 Excel Export Specification

- Admin clicks Export to Excel from dashboard
- System generates a .xlsx file matching the exact format of the existing tracker
- All colour coding, formulas, and conditional formatting preserved
- File is named: HCK_Intern_Tracker_[MonthYear].xlsx and downloaded immediately

5. Non-Functional Requirements

Category	Requirement
Performance	Dashboard loads within 2 seconds. Attendance grid handles 30 interns x 31 days without lag.
Availability	99% uptime during working hours (9am–7pm). Maintenance windows during off-hours only.
Mobile Responsive	Full functionality on mobile browser — interns check in and log calls from their phones.
Security	Role-based access enforced at API level. No intern can view another intern's data. HTTPS only.
Data Retention	Attendance and performance data retained for minimum 2 years. Exportable at any time.
Browser Support	Chrome, Firefox, Safari, Edge — latest 2 versions. No IE support required.
Offline Tolerance	If internet is unavailable during check-in, system queues the action and syncs when connection restores.
Audit Log	All attendance edits logged with: who changed it, what was changed, and when. Visible to admin only.

6. Recommended Technology Stack

The following stack is recommended based on the requirements. Final decisions are at the discretion of the development team.

Layer	Recommended	Notes
Frontend	React.js + Tailwind CSS	Fast, component-based, mobile-friendly
Backend / API	Node.js + Express OR Django (Python)	Python preferred if dev team is familiar — easier Excel parsing with openpyxl
Database	PostgreSQL	Relational — handles attendance grid and leave logic well
Excel Parsing	openpyxl (Python) or SheetJS (JS)	openpyxl recommended — same library used in current Excel build
Authentication	JWT tokens + bcrypt	Secure, stateless, easy role management
Notifications	Email via SendGrid + Browser Push (Web Push API)	For leave approvals and KPI alerts
Hosting	Heroku / Railway / DigitalOcean (VPS)	Budget-friendly options for Nepal-based deployment
File Storage	Local server or AWS S3	For uploaded Excel files and exported reports

7. Key Screens & Features Summary

Screen	Visible To	Key Features
Login Page	All	Email/password, role auto-detected, forgot password
Intern Dashboard	Intern	Today's status, CL balance, call log form, tour log form, leave apply button, own monthly stats
Check-In Page	Intern	One-tap check-in, time stamp shown, late warning if past shift time
Call Log Form	Intern / Supervisor	Date, calls made, connected, visits booked, registrations, notes
Tour Log Form	Intern / Supervisor	Date, visitor name, count, outcome, notes
Leave Application	Intern	Date picker, leave type dropdown, reason text, CL balance shown if CL selected
Team Attendance Grid	Supervisor / Admin	Monthly grid, dropdown per cell, colour-coded, row totals, comp leave columns
Manager Dashboard	Admin	Daily summary panel, leaderboard, flags, pending approvals, alerts
Intern Profile	Admin	All profile fields, shift settings, Saturday roster, KPI targets
Leave Approval	Supervisor / Admin	List of pending requests, approve/reject with note, auto-updates grid
Reports & Export	Admin	Monthly summary table, export to .xlsx, date range filter, intern filter
Excel Sync Panel	Admin	Upload Excel, preview diff, confirm import, export current data to Excel
Audit Log	Admin	All edits with timestamp, user, before/after values

8. Development Phases

Phase	Name	Deliverables	Est. Duration
Phase 1	Core Foundation	Authentication, intern profiles, attendance grid (web), basic dashboard, check-in system	3–4 weeks
Phase 2	Performance & Leave	Call log, tour log, KPI targets, leave application and approval workflow, CL auto-logic	3–4 weeks
Phase 3	Excel Integration	Excel upload parser, conflict diff viewer, Excel export with formatting preserved, audit log	2–3 weeks
Phase 4	Alerts & Polish	Email and browser notifications, leave pattern detection, mobile responsiveness, performance flags, UAT and bug fixes	2 weeks

Total estimated development time: 10–13 weeks with a dedicated developer. Can be reduced to 6–8 weeks with a 2-person team.

9. Constraints & Assumptions

9.1 Constraints

- The system must remain compatible with the existing Excel tracker format — no breaking changes to the Excel structure
- All intern data is confidential — must not be accessible outside the college's authorised users
- System must work on basic Android smartphones (interns' primary devices for check-in)
- Internet connectivity in the Herald College building is assumed — no offline-first architecture required in Phase 1

9.2 Assumptions

- Intern IDs (INT-01 to INT-30) will be the primary key linking Excel data to web profiles
- Saturday duty rosters are set in advance by admin at the start of each month
- Interns have personal smartphones with internet access for self check-in
- One supervisor is assigned per group of 8–10 interns
- The development team will be provided access to the current Excel file for reference

10. Document Approval

This document requires review and sign-off before development begins.

Name	Role	Signature	Date
	Manager / Admin		
	Development Lead		
	Project Sponsor		

END OF DOCUMENT — InternTrack SRD v1.0 — Herald College Kathmandu